Université Paris 1 Panthéon-Sorbonne
Master 2 Finance, Technology & Data

Final Exam Submission

Counterfactual Inflation Analysis

Ukraine in the Euro Area

Evelyn FAGBOLA

Course: Quantitative Methods in Finance

Instructor: Eric Vansteenberghe

1 Part A. Ukraine's Monetary Regime: A Preliminary Analysis (2000–2025)

1.1 Summary Table

Period	Approximate UAH/USD rate	De facto regime classification	Key events
Jan 2000 - Mar 2005	5.33	De facto peg (USD)	At the beginning of the year 2000, the NBU officially declared a free float but ended up regularly intervening in order to keep the exchange rate around UAH 5.33 per USD. Monetary policy at this time was used to keep the exchange rate stable and maintain the peg. (1)
Mar 2005 - Apr 2008	5.00–5.06	De facto peg (USD)	The NBU allowed the interbank exchange rate to fluctuate in a narrow range of UAH 5.00 - 5.06 per USD. However, the regime is still a de facto peg. (1)
Apr 2008 - Feb 2009	5.05 to 7.88	Managed exchange rate and peg breakdown	From March 2008, the NBU allowed the exchange rate to rise above the previous de facto range. Then the global financial crisis that occured during this time cause the peg to collapse. An IMF SBA (Stand By Arrangement) was approved in November 2008 and the IMF retroactively reclassified the regime from April 2008. (2)
Feb 2009 - Feb 2014	7.7 - 8.0	De facto peg (USD)	The exchange rate was effectively re-pegged to the US dollar but the crisis restrictions were not fully removed under the 2010 SBA. There IMF 2012 Article IV confirms a stablised arrangement from 1 March 2010. (3)(4)
Feb 2014 - Sep 2014	8.0 to 12–13	Forced float	Due to increasing pressure on the hryvnia and the critical reduction of the reserves, the NBU allowed the exchange rate to float in February 2014. Then in April 2014, the hryvnia depreciated by more than 35% due to the annexation of the Crimea. (5)(8)
Sep 2014 - Mar 2015	13 to 20–25	Managed float	This period is the second half of the crisis depreciation. Formally, the exchange rate remained floating but from September 2014 to March 2015 the NBU increased their crisis management. This was done through several price FX auctions, tighter intervention rules, a derivative ban and limitations on banks' FX positions. During 2014, the hryvnia fell by around 90%. Inflation reach 25% at the end of 2014 and 43% at the end of 2015. (6)(9)
Mar 2015 - Sep 2016	30.0 peak 23.5 end-2015 27.2 end-2016	Managed float with transition to inflation targeting	Following the exchange rate crisis, the NBU adopted a formally floating regime allowing the hryvnia to stabilise. Importantly, this period was still marked by extensive administrative FX restrictions. Policy focused on tight monetary control, rebuilding reserves and a gradual move toward inflation targeting. By mid 2016 the NBU progressively removed the emergency measures introduced in 2014 and 2015. (7)(8)(9)(10)

Period	Approximate UAH/USD rate	De facto regime classification	Key events
Sep 2016 - Feb 2019	27	Floating and inflation targeting	During this period, there was a focus on normalisation in inflation targeting. Despite the IMF classification of the exchange rate as floating from 2014 onward, capital controls and FX restrictions still remained. The NBU formally adopted inflation targeting in December 2016 and announced a declining target path from 12% in 2016 to 5% from 2019 onward. Over this period, the exchange rate was described as floating and intervention was limited to smoothing excess volatility and reubuilding reserves, not defending a peg. Anti-crisis FX restrictions were progressively rolled back. This was the most evident period of monetary autonomy in the sample. (10)(11)(12)(22)(23)
Feb 2019 - Dec 2021	Mid/High 20s to 27.3 end-2021	Floating and inflation targeting	Following the new FX regime which took effect on 7th February 2019, the NBU maintained a floating exchange rate within the inflation targeting framework. There is a continued unwinding of the 2014 and 2015 anti-crisis restrictions and the FX liberalisation continued through 2021. This included easing rules on FX purchases, derivatives and individuals foreign investments. In this period the exchange rate remained market-determined and interventions were again only used to smooth volatility and build reserves. (13)(21)(22)(23)
Dec 2021 - Oct 2023	29.25 - fixed at start of the invasion 36.57 - adjusted on 21 July 2022	Wartime fixed exchange rate	After the full-scale invasion of Ukraine, a wartime emergency exchange rate was introduced. Then on 24 February 2022, the NBU, aiming to preserve reserves, fixed the official exchange rate, limited cash withdrawals, maintained cashless payments, imposed FX restrictions and capital controls. This also aided in containing panic. In July 2022, after major wartime changes to economic fundamentals, the NBU devalued the official rate by 25% to 36.57. This was not a return to peacetime inflation targeting or to the market float but it was a wartime stabilisation regime. This was a based on a fixed nominal anchor and extensive administrative controls aimed at keeping the financial system functioning under martial law. At the end of the period, the NBU moved to managed exchange rate flexibility. (14)(15)(16)(17)

Period	Approximate UAH/USD rate	De facto regime classification	Key events
Oct 2023 - Dec 2025	42.04	Managed float and flexible inflation targeting	On the 3rd of October 2023, the NBU ended the wartime fixed-rate regime and shifted to managed flexibility. From then on, the official exchange rate was determined by interbank transactions rather than being set administratively. The IMF reclassified the de facto arrangement from stabilized to floating with effect from that 3rd October. In 2024, the NBU stated that it had transitioned to flexible inflation targeting as an intermediate regime as they worked their way to full-fledged inflation targeting with a floating exchange rate. The exchange rate was allowed to move in both directions but the NBU continued to cover the structural FX deficit, smooth excess volatility, and preserve reserves. By 2025 the NBU officially described the current framework as flexible inflation targeting supported by managed flexibility and still-existing FX restrictions. (18)(19)(20)

Note: Citations are presented in brackets and correspond to entries in the references section. These citations pertain to both approximate exchange rate and key events columns.

1.2 Brief Argument

During most of this period, Ukraine did not have a genuine monetary policy sovereignty. This sovereignty is seen mainly in periods without a de facto dollar peg such as from September 2016 - December 2021 when the NBU formally adopted inflation targeting, with a floating exchange rate and intervention was used to smooth volatility rather than to defend a peg. Some sovereignty did start to appear after the collapse of the peg in 2014 but the period of 2014-2016 was still marked by crisis management. After October 2023, sovereignty started to emerge again under managed flexibility and flexible inflation target, but it was still limited as the NBU continued to manage the FX market closely and maintain restrictions. By contrast, from January 2000 - April 2008 and February 2009 - February 2014, the monetary policy de facto was constrained by the exchange rate peg. In both periods, the hryvnia was effectively tied to the US dollar, so monetary policy was used to maintain exchange-rate stability rather than for domestic stabilisation goals. Importantly, the wartime period from February 2022 to October 2023 also greatly limited their monetary sovereignty even though it was a period of wartime stabilisation rather than a normal prewar peg. This information implies that the Euro Area membership would not represent the same treatment for the whole sample. In the long de facto dollar peg periods, joining the euro would have not indicated a massive additional loss of sovereignty because the NBU had already sacrificed much of its effective monetary independence during these periods. The biggest treatment effect would be in periods when Ukraine had opportunities to run its own monetary policy, especially from September 2016 - December 2021 when inflation targeting and the floating exchange rate gave the NBU greater sovereignty. Thus, the counterfactual impact of the Euro Area membership would be more important in the period after 2016 compared to earlier peg periods.

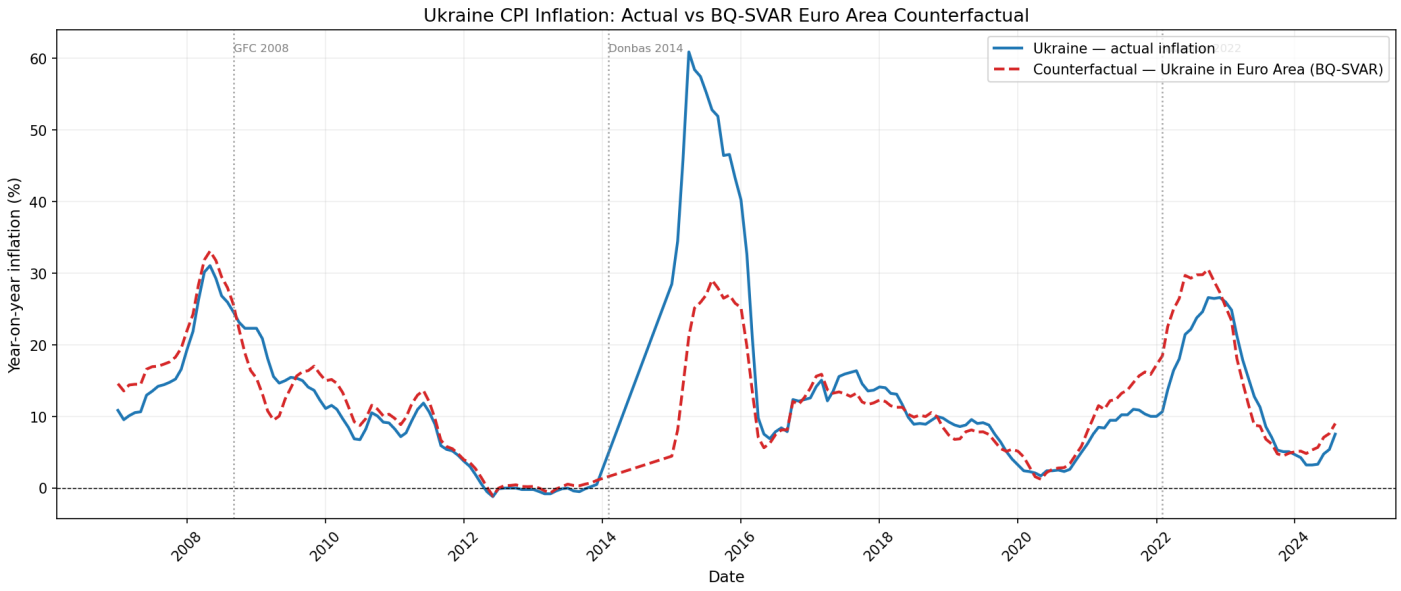

Figure 1: BQ-SVAR Ukraine in Euro Area Counterfactual. Rather than averaging euro area inflation, this approach feeds EA demand shocks through Ukraine's estimated VAR, preserving country-specific transmission.

2 Part B. Counterfactual Inflation Analysis

The BQ-SVAR counterfactual implies that the costs and benefits of monetary sovereignty changed considerably over the three crisis periods. In the 2008–2009 period, counterfactual inflation closely tracks Ukraine's actual inflation, suggesting that no large inflation cost or benefit resulted from monetary sovereignty. As Ukraine was already under a hard-dollar peg, it lacked effective monetary sovereignty so Euro Area membership would have had little impact on inflation dynamics. In the 2014–2015 period, actual inflation stood at 60%, while counterfactual inflation was around 29%. This difference reflects the collapse of the hryvnia, which pushed domestic prices to rise sharply through import cost inflation. This channel would not have existed if Ukraine held the euro since the exchange rate would have been fixed. Thus, Euro Area membership would have prevented this devaluation-driven inflation spike. In 2022, counterfactual inflation rises above actual inflation, meaning that monetary sovereignty was beneficial, as the NBU could contain inflation through exchange-rate stabilisation. However, under Euro Area membership, the European energy shock would have passed more directly into Ukrainian inflation, as Ukraine would have lacked the autonomy to offset it through exchange-rate and liquidity policies. Overall, the counterfactual does not simply imply that Ukraine would have experienced lower inflation had it been inside the Euro Area. Without its own central bank to provide emergency liquidity, Ukraine would have lost an important institutional tool on which it relied upon under martial law. Thus, it might have avoided a currency crisis while becoming more exposed to financing and debt-sustainability pressures in the future. The true cost of monetary sovereignty depends on the kind of macroeconomic crisis to which a country is most exposed. For Ukraine, retaining monetary sovereignty meant absorbing inflation spikes caused by devaluation, while preserving the emergency policy flexibility that became crucial in 2022.

3 References

1. International Monetary Fund (2010). *Ukraine - Request for Stand: By Arrangement and Cancellation of Current Arrangement Staff Report; Staff Supplement; Press Release on the Executive Board Discussion*. IMF Staff Country Reports, [online] 10(262). doi:https://doi.org/10.5089/9781455206841.002.

2. International Monetary Fund (2011). *Ukraine: Ex Post Evaluation of Exceptional Access Under the 2008 Stand-By Arrangement*. IMF Staff Country Reports, [online] 11(325). Available at: https://www.imf.org/external/pubs/ft/scr/2011/cr11325.pdf.

3. International Monetary Fund (2012). *Ukraine: Staff Report for the 2012 Article IV Consultation—Informational Annex*. IMF Staff Country Reports, [online] 2012(315). doi:https://doi.org/10.5089/9781475591347.002.A002.

4. International Monetary Fund (2014). *Ex Post Evaluation of Exceptional Access Under the 2010 Stand-By Arrangement*. IMF Staff Country Reports, [online] 14(146). Available at: https://www.imf.org/external/pubs/ft/scr/2014/cr14146.pdf

5. International Monetary Fund (2014). *Ukraine: Request for a Stand-By Arrangement—Staff Report; Supplement; Staff Statement; Press Release; and Statement by the Executive Director for Ukraine*. IMF Country Report 14(106). [online] Available at: https://www.imf.org/external/pubs/ft/scr/2014/cr14106.pdf

6. International Monetary Fund (2015). *Annual Report on Exchange Arrangements and Exchange Restrictions*. IMF. [online]. Available at: https://www.imf.org/-/media/files/publications/areaer/areaer_2015_overview.pdf

7. The National Bank of Ukraine (2015). *Monetary Policy Fundamentals*. [online]. Available at: https://bank.gov.ua/admin_uploads/article/MPG_2015_eng.pdf?v=7

8. International Monetary Fund (2016). *Ex Post Evaluation of Exceptional Access Under the 2010 Stand-By Arrangement*. IMF Staff Country Reports, [online] 14(146). Available at: https://www.imf.org/external/pubs/ft/scr/2014/cr14146.pdf

9. National Bank of Ukraine (2016). *National Bank of Ukraine Push ahead with Easing FX Regulation and Capital Controls*. [online] National Bank of Ukraine. Available at: https://bank.gov.ua/en/archive-news/all/32278864-national-bank-of-ukraine-push-ahead-with-easing-fx-regulation-and-capital-controls

10. International Monetary Fund (2017). *IMF Executive Board Concludes 2016 Article IV Consultation with Ukraine*. [online] Available at: https://www.imf.org/en/news/articles/2017/04/04/pr17112-imf-executive-board-concludes-2016-article-iv-consultation-with-ukraine

11. International Monetary Fund (2017). *Ukraine - Selected Issues*. IMF Country Report, [online] 17(84). Available at: https://www.imf.org/-/media/files/publications/cr/2017/cr1784.pdf

12. National Bank of Ukraine (2017). Inflation report. [online] Available at: https://bank.gov.ua/admin_uploads/article/IR_2017-Q2_eng.pdf

13. National Bank of Ukraine (2021). *Inflation Report*. [online] Available at: https://bank.gov.ua/admin_uploads/article/IR_2021-Q3_eng.pdf

14. Board of the National Bank of Ukraine (2022). On Operation of Banking System Under Martial Law. [online]. Available at: https://bank.gov.ua/admin_uploads/law/Resolution_24022022_18_kp_eng.pdf

15. International Monetary Fund (2022). *Request For Purchase Under The Rapid Financing Instrument And Cancellation Of Stand-By Arrangement—Press Release; Staff Report; And Statement By The Executive Director For Ukraine*. IMF Country Report, [online] 22(74). Available at: https://www.imf.org/-/media/files/publications/cr/2022/english/1ukrea2022001.pdf

16. National Bank of Ukraine (2022). *FX restrictions and exchange rate policy*. [online]. Available at: https://bank.gov.ua/en/markets/liberalization

17. National Bank of Ukraine (2023). *NBU 2022 Annual Report: Safeguarding Macrofinnancial Stability and Public Confidence despite Unprecedented Wartime Challenges*. [online] National Bank of Ukraine. Available at: https://bank.gov.ua/en/news/all/richniy-zvit-nbu-za-2022-rik-zbereje nnya-makrofinansovoyi-stabilnosti-ta-doviri-gromadyan-popri--bezpretsedentni-v iprobuvannya-viynoyu

18. National Bank of Ukraine (2023). *NBU Introduces Managed Flexibility of Exchange Rate to Strengthen Resilience of the economy and the FX Market*. [online] National Bank of Ukraine. Available at: https://bank.gov.ua/en/news/all/nbu-vprovadzjuye-kerovanu-gnuchkist-obminnogo-kursu-s cho-posilit-stiykist-valyutnogo-rinku-ta-ekonomiki

19. International Monetary Fund (2024). *Fourth Review Of The Extended Arrangement Under The Extended Fund Facility, Request For Modification Of A Performance Criterion, And Financing Assurances Review—Press Release; Staff Report; And Statement By The Executive Director For Ukraine*. [online] National Bank of Ukraine. Available at: https://www.imf.org/-/media/files/publications/c r/2024/english/1ukrea2024002-print-pdf.pdf

20. National Bank of Ukraine (2024). Annual Report 2024. [online]. Available at: https://bank.gov.u a/admin_uploads/article/annual_report_2024_eng.pdf

21. International Monetary Fund (2025). *IMF Executive Board Completes the Eighth Review of the Extended Arrangement under the Extended Fund Facility for Ukraine*. [online] Available at: https://www.imf.org/en/news/articles/2025/06/30/pr-25227-ukraine-imf-completes-8th-rev-of-ext-a rrang-under-eff

22. 2National Bank of Ukraine (2026). *About Monetary Policy*. [online] Available at: https://bank.gov .ua/en/monetary/about

23. National Bank of Ukraine (2026). *Currency Liberalization*. [online] Available at: https://bank.gov .ua/en/markets/liberalization/old-liberalization